Chapter 1: The Real and Complex Number Systems

Section: The Complex Field

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Definition and Field Structure

The Real Numbers Inside $\mathbb{C}$

Conjugation and Absolute Value

Properties of the Absolute Value

Summation Notation and the Schwarz Inequality

Summary

Definition and Field Structure

Why the Complex Numbers?

- We have built $\mathbb{R}$ with the least upper bound property

Why the Complex Numbers?

- We have built $\mathbb{R}$ with the least upper bound property
- But not every polynomial has a root in $\mathbb{R}$ — e.g., $x^2 + 1 = 0$

Why the Complex Numbers?

- We have built $\mathbb{R}$ with the least upper bound property
- But not every polynomial has a root in $\mathbb{R}$ — e.g., $x^2 + 1 = 0$
- Solution: extend $\mathbb{R}$ to the complex numbers $\mathbb{C}$

Definition 1.24: Complex Number

Definition 1.24

A complex number is an ordered pair (a, b) of real numbers.

Equality: $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Definition 1.24: Arithmetic Operations

Let $x = (a, b)$ and $y = (c, d)$.

Addition:

$$x + y = (a + c,\ b + d)$$

Definition 1.24: Arithmetic Operations

Let $x = (a, b)$ and $y = (c, d)$.

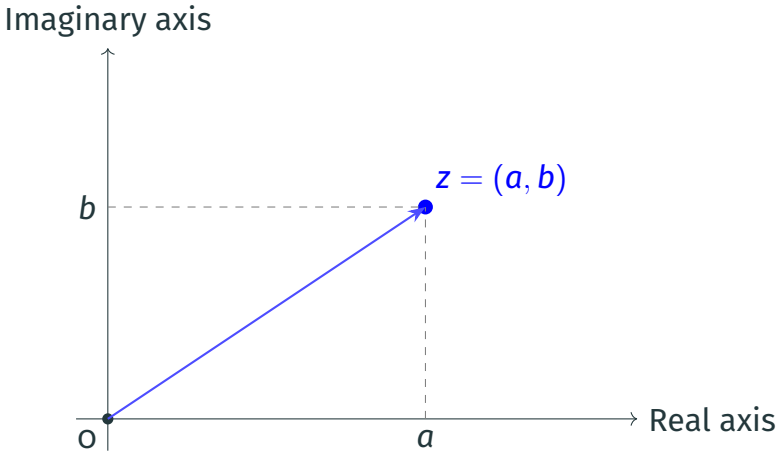
Addition:

$$x + y = (a + c,\ b + d)$$

Multiplication:

$$xy = (ac - bd,\ ad + bc)$$

The Complex Plane



Theorem 1.25: $\mathbb{C}$ Is a Field

Theorem 1.25

The set of all complex numbers, with the addition and multiplication of Definition 1.24, is a field.

The additive identity is $(0, 0)$ and the multiplicative identity is $(1, 0)$.

Proof Strategy: Theorem 1.25

Goal: Verify all field axioms from Definition 1.12.

Approach: Let $x = (a, b)$, $y = (c, d)$, $z = (e, f)$.

(A1)–(A5) Addition: closure, commutativity, associativity, identity $(0, 0)$, inverse $(-a, -b)$

(M1)–(M5) Multiplication: closure, commutativity, associativity, identity $(1, 0)$, inverse $\frac{1}{x}$

(D) Distributive law

Proof of Theorem 1.25 — Addition Axioms

Addition Axioms (A1)–(A5)

(A2): $x + y = (a + c,\ b + d) = (c + a,\ d + b) = y + x$

(A3): $(x + y) + z = (a + c + e,\ b + d + f) = x + (y + z)$

(A4): $x + (0, 0) = (a, b) = x$

(A5): Set $-x = (-a, -b)$. Then $x + (-x) = (0, 0)$

Proof of Theorem 1.25 — Multiplication Axioms

(M2), (M3), (M4)

(M2): $xy = (ac - bd,\ ad + bc) = (ca - db,\ da + cb) = yx$

(M3):

$$\begin{aligned}(xy)z &= (ac-bd,\ ad+bc)(e,f)\\ &= (ace-bde-adf-bcf,\ \ acf-bdf+ade+bce)\\ &= (a,b)(ce-df,\ cf+de)\\ &= x(yz)\end{aligned}$$

(M4): $(1,0)(a,b) = (a,b) = x$

Proof of Theorem 1.25 — Multiplicative Inverse

Inverse (M5)

If $x = (a, b) \neq (0, 0)$, then $a^2 + b^2 > 0$ (by Proposition 1.18).

Define:

$$\frac{1}{x} = \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right)$$

Proof of Theorem 1.25 — Multiplicative Inverse

Inverse (M5) — Verification

Set $D = a^2 + b^2$. Then:

$$x \cdot \frac{1}{x} = (a,\ b)\left(\frac{a}{D},\ \frac{-b}{D}\right)$$

$$= \left(\frac{a^2}{D} + \frac{b^2}{D},\ \frac{-ab}{D} + \frac{ab}{D}\right) = \left(\frac{D}{D},\ 0\right) = (1, 0)$$

Proof of Theorem 1.25 — Distributive Law

Distributivity (D)

$$
\begin{aligned}
x(y+z) = (a,b)(c+e,\ d+f) \\
&= (a(c+e) - b(d+f),\ a(d+f) + b(c+e)) \\
&= (ac - bd,\ ad + bc) + (ae - bf,\ af + be) \\
&= xy + xz
\end{aligned}
$$

The Real Numbers Inside $\mathbb{C}$

Theorem 1.26: Embedding $\mathbb{R}$ into $\mathbb{C}$

Theorem 1.26

For any real numbers a and b:

$$(a, 0) + (b, 0) = (a + b, 0), \qquad (a, 0)(b, 0) = (ab, 0).$$

Key consequence: We identify $(a, 0)$ with a.

$\implies \mathbb{R}$ is a *subfield* of $\mathbb{C}$.

Definition 1.27: The Imaginary Unit

Definition 1.27

$$i = (0, 1)$$

Theorem 1.28: $i^2 = -1$

Theorem 1.28

$i^2 = -1.$

Proof

$i^2 = (0,1)(0,1) = (0 \cdot 0 - 1 \cdot 1,\ 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1.$

Theorem 1.29: The $a + bi$ Notation

Theorem 1.29

If a and b are real, then $(a, b) = a + bi$.

Proof

$$a + bi = (a, 0) + (b, 0)(0, 1) = (a, 0) + (0, b) = (a, b).$$

Conjugation and Absolute Value

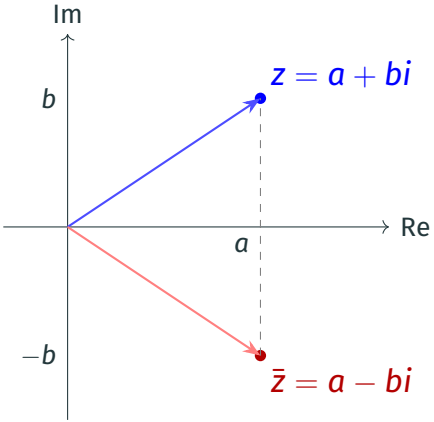
Definition 1.30: Conjugate, Real and Imaginary Parts

Definition 1.30

If $z = a + bi$ with $a, b \in \mathbb{R}$, then:

- The conjugate of z is $\bar{z} = a - bi$
- $a = \mathrm{Re}(z)$ is the *real part* of z
- $b = \mathrm{Im}(z)$ is the *imaginary part* of z

Conjugation in the Complex Plane



Theorem 1.31: Properties of Conjugation

Theorem 1.31

If z and w are complex, then:

(a) $\overline{z+w} = \bar{z} + \bar{w}$

Theorem 1.31: Properties of Conjugation

Theorem 1.31

If z and w are complex, then:

(a) $\overline{z+w} = \bar{z} + \bar{w}$

(b) $\overline{zw} = \bar{z} \cdot \bar{w}$

Theorem 1.31: Properties of Conjugation

Theorem 1.31

If z and w are complex, then:

(a) $\overline{z+w} = \bar{z} + \bar{w}$

(b) $\overline{zw} = \bar{z} \cdot \bar{w}$

(c) $z + \bar{z} = 2\,\mathrm{Re}(z), \quad z - \bar{z} = 2i\,\mathrm{Im}(z)$

Theorem 1.31: Properties of Conjugation

Theorem 1.31

If z and w are complex, then:

(a) $\overline{z+w} = \bar{z} + \bar{w}$

(b) $\overline{zw} = \bar{z} \cdot \bar{w}$

(c) $z + \bar{z} = 2\,\mathrm{Re}(z), \quad z - \bar{z} = 2i\,\mathrm{Im}(z)$

(d) $z\bar{z}$ is real and positive (except when $z = 0$)

Key fact: $z\bar{z} = a^2 + b^2$

Definition 1.32: Absolute Value

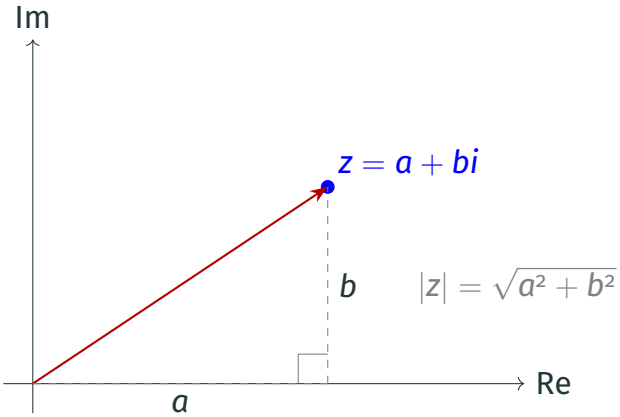
Definition 1.32

If z is a complex number, its absolute value is

$$|z| = (z\bar{z})^{1/2} = \sqrt{a^2 + b^2}$$

Note: When x is real: $|x| = \sqrt{x^2}$, so $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

Absolute Value in the Complex Plane



Properties of the Absolute Value

Theorem 1.33: Properties of Absolute Value

Theorem 1.33

Let z and w be complex numbers. Then:

(a) $|z| > 0$ unless $z = 0$; $|0| = 0$

Theorem 1.33: Properties of Absolute Value

Theorem 1.33

Let z and w be complex numbers. Then:

(a) $|z| > 0$ unless $z = 0$; $|0| = 0$

(b) $|\bar{z}| = |z|$

Theorem 1.33: Properties of Absolute Value

Theorem 1.33

Let z and w be complex numbers. Then:

(a) $|z| > 0$ unless $z = 0$; $|0| = 0$

(b) $|\bar{z}| = |z|$

(c) $|zw| = |z|\,|w|$

Theorem 1.33: Properties of Absolute Value

Theorem 1.33

Let z and w be complex numbers. Then:

(a) $|z| > 0$ unless $z = 0$; $|0| = 0$

(b) $|\bar{z}| = |z|$

(c) $|zw| = |z|\,|w|$

(d) $|\operatorname{Re} z| \leq |z|$

Theorem 1.33: Properties of Absolute Value

Theorem 1.33

Let z and w be complex numbers. Then:

(a) $|z| > 0$ unless $z = 0$; $|0| = 0$

(b) $|\bar{z}| = |z|$

(c) $|zw| = |z|\,|w|$

(d) $|\operatorname{Re} z| \leq |z|$

(e) $|z + w| \leq |z| + |w|$ (triangle inequality)

Proof of Theorem 1.33(c): Multiplicativity

Proof of (c)

Write $z = a + bi$, $w = c + di$. Then:

$$|zw|^2 = (ac - bd)^2 + (ad + bc)^2 = (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2$$

So $|zw|^2 = (|z|\,|w|)^2$, and both sides are nonneg. $\Longrightarrow |zw| = |z|\,|w|$.

Proof of Theorem 1.33(e): Triangle Inequality — Setup

Proof of (e) — Key identity

Note: $\bar{z}w$ is the conjugate of $z\bar{w}$, so

$$z\bar{w} + \bar{z}w = 2\,\mathrm{Re}(z\bar{w})$$

Expand:

$$\begin{aligned} |z+w|^2 &= (z+w)\overline{(z+w)} = (z+w)(\bar{z}+\bar{w}) \\ &= z\bar{z} + z\bar{w} + \bar{z}w + w\bar{w} \\ &= |z|^2 + 2\,\mathrm{Re}(z\bar{w}) + |w|^2 \end{aligned}$$

Proof of Theorem 1.33(e): Triangle Inequality — Conclusion

Proof of (e) — Completing the estimate

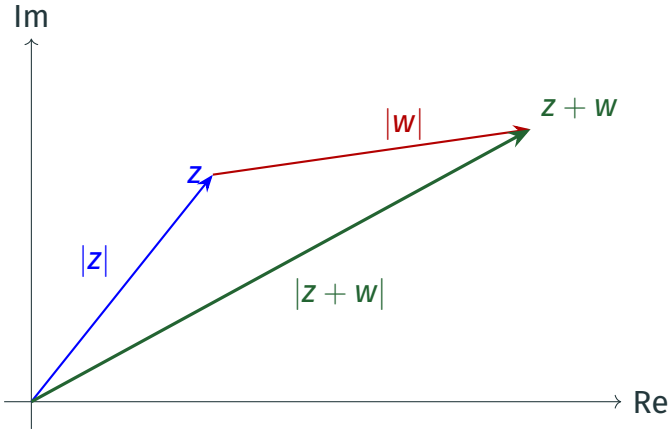
By part (d): $\mathrm{Re}(z\bar{w}) \leq |z\bar{w}|$. By part (c): $|z\bar{w}| = |z|\,|w|$.

Therefore:

$$|z+w|^2 \leq |z|^2 + 2|z|\,|w| + |w|^2 = \big(|z| + |w|\big)^2$$

Taking square roots: $\boxed{|z+w| \leq |z| + |w|}$

Triangle Inequality: Geometric View



$$|z + w| \leq |z| + |w|$$

Summation Notation and the Schwarz Inequality

Notation 1.34: Summation

Notation 1.34

If $x_1, \ldots, x_n$ are complex numbers, we write

$$x_1 + x_2 + \cdots + x_n = \sum_{j=1}^{n} x_j.$$

Theorem 1.35: The Schwarz Inequality

Theorem 1.35 (Schwarz Inequality)

If $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ are complex numbers, then

$$\left| \sum_{j=1}^{n} a_j \bar{b}_j \right|^2 \leq \sum_{j=1}^{n} |a_j|^2 \sum_{j=1}^{n} |b_j|^2.$$

Proof of Theorem 1.35 — Setup

Setup

Write:

$$A = \sum |a_j|^2, \qquad B = \sum |b_j|^2, \qquad C = \sum a_j \bar{b}_j.$$

Trivial case: If $B = 0$, then $b_1 = \cdots = b_n = 0$ and the inequality holds.

Assume $B > 0$.

Proof of Theorem 1.35 — Key Computation

The trick

Expand $\sum |Ba_j - Cb_j|^2$:

$$\begin{aligned}\sum |Ba_j - Cb_j|^2 &= B^2 \sum |a_j|^2 - B\bar{C} \sum a_j \bar{b}_j - BC \sum \bar{a}_j b_j + |C|^2 \sum |b_j|^2 \\ &= B^2 A - B|C|^2 \\ &= B(AB - |C|^2)\end{aligned}$$

Proof of Theorem 1.35 — Conclusion

Conclusion

Each $|Ba_j - Cb_j|^2 \geq 0$, so

$$B(AB - |C|^2) \geq 0.$$

Since $B > 0$, divide both sides by B:

$$\boxed{AB - |C|^2 \geq 0 \qquad \Longleftrightarrow \qquad |C|^2 \leq AB}$$

Summary

Summary

Key takeaways from this section:

- Complex numbers are ordered pairs (a, b) of reals, forming a field $\mathbb{C}$

Summary

Key takeaways from this section:

- Complex numbers are ordered pairs (a, b) of reals, forming a field $\mathbb{C}$
- $\mathbb{R} \subset \mathbb{C}$ via $(a, 0) \leftrightarrow a$; $i = (0, 1)$ with $i^2 = -1$

Summary

Key takeaways from this section:

- Complex numbers are ordered pairs (a, b) of reals, forming a field $\mathbb{C}$
- $\mathbb{R} \subset \mathbb{C}$ via $(a, 0) \leftrightarrow a$; $\quad i = (0, 1)$ with $i^2 = -1$
- Conjugation ($\bar{z} = a - bi$) and absolute value ($|z| = \sqrt{a^2 + b^2}$) satisfy the triangle inequality

Summary

Key takeaways from this section:

- Complex numbers are ordered pairs (a, b) of reals, forming a field $\mathbb{C}$
- $\mathbb{R} \subset \mathbb{C}$ via $(a, 0) \leftrightarrow a$; $i = (0, 1)$ with $i^2 = -1$
- Conjugation ($\bar{z} = a - bi$) and absolute value ($|z| = \sqrt{a^2 + b^2}$) satisfy the triangle inequality
- The Schwarz inequality: $\left|\sum a_j \bar{b}_j\right|^2 \leq \sum |a_j|^2 \sum |b_j|^2$

Coming up next: Euclidean Spaces — generalizing $\mathbb{C}$ to $\mathbb{R}^k$.